

# 2025 Ch H1 Q24

Section: Chemistry in Society

Topic: Experimental Techniques

## Question summary

When carrying out a titration to determine the mass of vitamin C in a tablet, which of the following will not help determine the end-point accurately?

## Worked solution

Accurate end-point detection depends on good visual contrast and controlled addition of titrant near the end-point.

Adding solution dropwise near the end-point improves accuracy (helps).

A white tile under the flask improves contrast to see the colour change (helps).

Swirling the flask ensures thorough mixing and a sharp end-point (helps).

Using a dry conical flask is unnecessary for a titration—the conical flask may be rinsed with water without affecting the moles of analyte (its volume is measured by pipette). Therefore, this does not help end-point accuracy.

## Final answer

**C. Using a dry conical flask**

## Revision tips

- Only the burette and pipette must be rinsed with the solutions they will contain; the conical flask can be rinsed with water.
- Add titrant dropwise near the end-point and swirl continuously for a clear, sharp colour change.
- Use a white background to enhance visibility of subtle colour changes.